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**ECONOMICS
MADE EASY**

ECONOMICS MADE EASY

BY

P V RAJEEV



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Economics Made Easy
P V Rajeev



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PREFACE

I have written this book for the benefit of ordinary readers who do not intend to specialize in Economics. Rather it is a book written for the layman so that he could become familiar with some of the basic concepts in Economics which will enable him to follow discussions in Economics as they appear in popular newspapers and magazines. The book may be of special interest to Engineers, Accountants, Managers, IT Professionals, Bankers, Businessmen, Civil Servants, Journalists, Political and Trade Union Workers and other professionals who would like to know some Economics but have no desire to specialize in the field. A new student intending to specialize in Economics may also find the book useful because it is so simple that it may help him in understanding the basic concepts of Economics more thoroughly. An advanced student of Economics may also enjoy reading the book because of his passion to read and digest everything that is written on Economics. It is for the readers to judge the value of the book.

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CHAPTER I

THE ECONOMY

A housewife in Delhi goes to the market to buy toothpaste, a bottle of shampoo, a packet of butter, some eggs, some vegetables and fruits. She is back home in half an hour having purchased all her requirements. The commodities she purchased had come from different parts of India but they were all available in the nearby market. What is that force which brought all our housewife's requirements to the nearby market? The Government of India had not issued directives that such and such commodities produced at such and such places should be sent to such and such markets. Yet this is exactly what has happened.

Our housewife is not able to cite even one example when she went to the market to buy toothpaste and had to return home without it because it was not available for sale. This force, which makes the housewife's requirements available in the nearby market was described as the 'invisible hand' by Adam Smith, the father of modern economics.

Another name that can be given to the 'invisible hand' is market forces. The producer who manufactured

toothpaste and green vegetables did so to meet his self interest of earning profits. The retailer who sold these items to our housewife also did so with the motive of earning an income. So also the process of transporting toothpaste and vegetables from their place of production to the retail market was carried out by people, each doing something out of their own self interest. Yet, what is a matter of self interest to so many people, is also in the interest of society as is reflected in the interest of our housewife and so many others like her.

The existence of an ‘invisible hand’ is beyond dispute. This also indicates that there are certain natural laws operating in the system which may be called the law of market forces. A great deal of the subject matter of economics relates to the study of the operation of these market forces, its imperfections and the remedial measures that are required to be taken to check the imperfect operation of market forces.

There are several elements operating in an economy. They include the consumers, the producers, the traders, the government, and so on. Basically, all activity in the economy is ultimately to meet the needs of the consumer or the public. The consumers utilize their income to purchase various goods and services in the market. They also enjoy a number of free or subsidized services provided to them by the government. The individuals or households who ‘act as the consumers earn their income by engaging themselves in some kind of productive economic activity. Part of the income earned by an individual or a household is spent in purchasing goods and services and some part of it is utilized for saving and investment.

A very dynamic element of our economy are the producers. The producers may be organized as big companies or as small scale manufacturing units. The prime activity of the producers is to produce goods or services which they can sell to the public or to the government at a price. The profits they earn out of these sales contribute their income. In general the health or well being of an economy can be judged from the prosperity of the productive class. When profits are high and producers are inclined to invest more-and more in their productive activities, it is a sign that the health of the economy is sound.

Production in an economy can mean production of goods as well as services. The production of motor cars, toothpaste, vegetables and cereals are production of goods. In addition, a number of services like banking, insurance, transport, trade, education and medical care are also produced and sold in the market.

A great deal of the activities in an economy takes place undirected through the operation of market forces. To meet the needs of consumers for products like toothpaste, shoes, television sets, washing machines and the like, the society can safely rely on the operation of market forces. But there are some items like defense, law and order, public roads, postal services in rural areas, education and medical care for the poor and so on, which will not be supplied in adequate quantities by the operation of market forces. In such areas direct intervention of the government to produce and provide goods and services becomes necessary.

The private sector will enter into a line of production only if it yields an adequate return by way of profits.

The private sector may be ready to invest in providing postal services in urban areas because there is scope for earning an adequate return. But it will find it unprofitable to set up a post office in a remote village whose needs can be adequately met only by a government. So also the private sector may come forward to set up educational institutions and hospitals for the rich who are capable of paying for the services provided to them. But if all hospitals and educational institutions are run by the private sector on market lines, the requirements of the poor will be largely neglected. Thus in addition to defence and maintenance of law and order, the government feels it necessary to directly invest in several other areas of production and public works.

Left to itself the market forces carry out a great deal of the activities in an economy quite efficiently. But on several occasions it becomes necessary for the government to interfere with the operation of the market forces to correct various distortions. For instance, in India, during normal times the production and distribution of sugar can be left to the free operation of market forces. But under certain circumstances, as when there is a shortage of sugar and sugar prices are rising, it may be necessary for the government to introduce rationing of sugar to meet the needs of the poorer sections of society. Rationing of sugar could mean distributing a certain quantity of sugar per head of population through the public distribution system at low or subsidized prices.

Thus the study of economics is a study of the activities in an economy. Many of these activities are more efficiently performed by the action of market forces. But

there are several instances where the regulation of the government over the operation of market forces becomes necessary. There are also areas where the operation of market forces will simply not deliver the goods. In these areas the government will have to intervene directly as a producer of goods and services.

CHAPTER II

FACTORS OF PRODUCTION

In economics we talk of four factors of production, namely, land, labor, capital and enterprise, which are responsible for all production in the economy and all income generated. The income from land is rent, while labor earns wages, capital earns interest and enterprise earns profits. All income generated in the economy falls under one of the following categories appropriately defined, namely, rent, wages, interest and profit. We can, in this chapter, have a closer look at the characteristics of the four factors of production and the income generated by each.

A basic character of land is that its supply is comparatively limited, though not limited in absolute sense. The total land area of a country is fixed in terms of its political boundaries. In modern times the land area of a country may be considered to be inclusive of the area of its neighboring seas coming under the country's exclusive economic zone. One can even talk of the rights of a country over certain areas of space or of the Arctic and Antarctic regions and some areas of the ocean where it has established a base.

While the land area of a country defined by its political boundaries is fixed, the land area available for cultivation or for a particular economic activity is variable. Land area previously used for agriculture may be converted into a residential area or acquired for setting up an industry. Dry wasteland may be converted into fertile agricultural land by suitable provision of irrigation facilities. Thus the supply of land is not absolutely fixed in an economic sense. But being comparatively limited in supply the price of land has a tendency to increase continuously in real terms over a period of time.

Urban land is generally more costly than rural or agricultural land. This is because in urban areas land is scarce and more intensively used and more productively exploited. Moreover, in urban areas the effective supply of land is increased by the construction of multi-storied apartments and buildings.

The labor force of a country may be seen as a subset of the total population of the country. The population of a country below 15 years of age and above 60 is generally not included as being part of the labor force of the country. Some other sections of the society who are mentally retarded or physically incapacitated will also have to be excluded from the labor force of the country. Similarly females who choose to remain outside the labor force will also have to be excluded. The student population of a country is also generally excluded from the labor force of the country.

The labor force of a country tends to expand when the population of a country is growing. But the expansion of

the size of labor force also depends on the age distribution of the population. For instance a country which has a high concentration of the population in the below 15 years category is going to experience a large increase in its labor force for some years in the future. This will have severe repercussions on the unemployment situation in the country.

The labor force of a country can broadly be classified as salary earners and wage earners. The salary earners are those with a fixed employment, earning a monthly salary and enjoying other privileges like leave with pay and so on. On the other hand, the wage earners may be considered to be those who do not have a fixed tenure of employment but work for daily wages, often in the agricultural or rural sector and are not entitled to privileges like security of employment, leave with pay and so on. The category of salary earners may be further classified as full time employees and part time employees.

The problem of unemployment as facing the above two categories of labor are also different. The degree of unemployment of wage earners depends on the number of man days of employment, available to them in a year. If they remain unemployed for a large number of days in a year the intensity of unemployment among them may be considered to be high. On the other hand, in the case of salary earners the intensity of unemployment may be seen to be related to the average length of waiting period for getting a suitable employment for the job seekers

The capital resources of a country comes from the savings made by the households, producers and government

in a country. The more the savings in a country, more is the available resources for investment excluding the resources that come from abroad. Thus a country that has a high rate of saving has the potential of developing faster

The savings generated by the households are generally deposited in various savings schemes and this leads to generation of financial assets. The savings deposited by the public are used by the public institutions to provide loans for industrial and other productive activities. The savings deposited in a bank, for instance, are used to give loans to industrialists, to the government, to the farmers and so on. The individual who deposited the savings is paid a rate of interest by the banks. The banks give loans at a still higher rate of interest. The difference between interest charged on loans and the interest paid to depositors constitute the earning of the financial institutions.

Sometimes the households instead of depositing their savings in a bank use their savings to buy shares of different companies. The basic difference between depositing one's savings in a bank and investing in shares is that while the former deposit yields a fixed rate of interest, the income from the latter is variable and is subject to an element of risk. However, the average returns on shares is generally higher than the returns on bank deposits to compensate for the risks assumed by the investors investing in shares.

The fourth factor of production, namely, enterprise, is a critical one. It is this factor that coordinates the activities of the other three factors. It is the duty of the entrepreneur to utilize land, labor and capital and set up

a production unit. The entrepreneur pays rent for land, wages to labor, and interest on capital borrowed and the surplus he earns over and above his costs constitute his profits. The profits earned by an entrepreneur is primarily a reward for his organizing capacity and the risks he bears.

It is useful in economics to distinguish between an innovative entrepreneur and an imitative entrepreneur. It is the former who ventures out in new directions and sets up business in challenging areas. The innovative entrepreneur makes use of challenging new discoveries in science and technology and brings out unprecedented innovations introducing new products in the market and gives a practical shape to the theoretical and applied innovations of scientists and technologists. The original initiative in a new field comes from the innovative entrepreneur. Being first in the field he is usually able to make a good fortune before the imitative entrepreneur also enters the market copying the original innovation. Once the original innovation has proved to be a success, there comes a number of imitative entrepreneurs to follow suit. The role of an innovator is highly critical from the point of view of the technological progress of a nation. He is a dynamic element in the process of economic development and growth of a nation.

CHAPTER III

PRICE DETERMINATION

Man cannot live without water. Water is one of the most useful things for man. Yet the price of water in terms of money is almost zero or negligible. On the other hand, the utility of gold is almost negligible. Yet the price of gold is very high in the market. Thus we can see that the price of a commodity does not depend upon how useful it is for man. On the other hand, in economics we learn that the price of a commodity is determined by the operation of the forces of demand and supply. According to this theory, the price of water is low because its supply is abundant and the price of gold is high because its supply is limited in relation to demand.

We shall now examine how the price of just one commodity – apples – is determined in the market. The market we are considering is the whole of India and we also assume there is no export or import of apples in India. Suppose the price of apples is ruling at Rs 50 per kg in 2017. Based on this ruling price of apples let us say the farmers decide to produce 100 crores of apples in 2017. Further let us assume that 2017 turns out to be a very bad year for crops and actual production of apples falls short of expectation by 25 crore and the

total production of apples in 2017 turns out to be only 75 crores. Now that there is a shortage of production of apples, the producers decide to sell apples at Rs 70 per kg. They do this on the understanding that at the price of Rs 70 per kg, the total demand for apples in India in 2017 will be 75 crores. Their assumption turns out to be true and at the end of the year it was realized that 75 crores of apples were produced and sold in India in India at a price of Rs 70 per kg.

The increase in price of apples gives an incentive to apple growers to produce more apples in 2016. In 2018 the production of apples turn out to be 125 crores. But when the production target of 125 crores materializes, it is realized that the entire output cannot be sold at a price of Rs 70 per kg. The price of apples comes down to Rs 40 per kg in 2018 which is the price at which the entire output can be sold. Thus the price of apples in 2018 rules at Rs 40 per kg.

From the above example we can infer the following: at a price of Rs 50 per kg the supply of apples is 100 crores while at a price of Rs 70 per kg the supply is 125 crores. At the same time the demand for apples is 100 crores at a price of Rs 50 per kg, 75 crores at a price of Rs per kg and 125 crores at a price of Rs 40 per kg.

The above example when depicted in graphical form gives an upward sloping supply curve and a downward sloping demand curve as shown below. It means that the supply of apples increases when its price goes up and its demand comes down when the price increases. The equilibrium price of apples at which both supply and demand will be equal during a normal year would

be Rs 50 per kg when 100 crores of apples are produced and sold in the whole of India

The determination of market prices is a complex affair and the example we have seen is a highly simplified one. For simplicity we assumed that the price of apples remained constant all over India throughout a particular year. In fact the price varies from locality to locality and from day to day. Moreover, the price of apples also vary depending on its quality. Yet a simplified example helps us to understand the basic forces at work in price determination.

In general we can see that the production of a commodity can be increased by increasing its price. When the price of wheat increases in the country the producers receive an incentive to increase the production of wheat. They may shift some land hitherto used for paddy cultivation to wheat cultivation when the price of wheat goes up relative to paddy.

If the government fixes the price of an essential commodity at a low level there is the danger that the production of this commodity will fall short of demand. Under such a condition the government will be forced to introduce rationing of this commodity. Otherwise a shortage of this commodity will emerge and evils like black marketing may also start.

When the government feels the need for restricting the consumption of a particular commodity like petrol, the objective can be partly achieved by increasing the price of the commodity either by imposing a high duty or by directly fixing its price at a higher level.

Consumption of harmful commodities like cigarettes and alcohol can also be curtailed by introducing a heavy tax on these commodities.

By manipulating on the price mechanism the government can channelize production into desirable directions. For instance, when it is felt that consumption of natural rubber should be encouraged as against consumption of synthetic rubber a higher rate of tax can be imposed on the latter thereby increasing its price and reducing its consumption.

We have in this chapter examined certain aspects relating to the operation of the price mechanism with a view to show how important is its role in society and how its operation in the economy can be manipulated.

CHAPTER IV

MARKET STRUCTURE

The operation of market forces brings out the best results when there is perfect competition in the economy. Perfect competition is a situation where there are very large number of producers producing the same product and the size of no firm is so large that it can exercise a dominating influence over the market. Under such a condition the competition between the firms is such that they need to manufacture their products at very competitive price and a high level of efficiency and productivity in the market prevails.

When perfect competition or conditions close to it prevail, there is very little need for the government to intervene in the market and competitive conditions themselves ensure the optimum levels of production and prices prevail in the market. In this book we accept these conclusions as valid without going into the mechanism of proving how perfect competition in the market leads to the ideal of optimum price and production.

While on the one side of the market structure spectrum we have perfect competition, on the other side we have absolute monopoly. Absolute monopoly is that

condition where there is only one producer in the entire market and this firm enjoys a monopoly in determining its level of output and price. It can be proved that under conditions of absolute monopoly the level of production will be lower than the optimum and the price prevailing in the market will be higher than the optimum. Thus under monopoly the consumer tends to lose both in terms of lower production as well as higher prices. As against this, the amount of profits earned by the firm would be above what the firms would have earned, had perfect competition prevailed in the market.

The conditions of monopoly can be realized in the private sector as well as the public sector. The Indian Railways, for instance, is a public sector monopoly and the private sector is not allowed to invest in this field. When absolute monopoly prevails in the market, it is possible for the firm to fix its output and prices at a level which a firm desires. If the monopoly is a private sector enterprise it will fix a level of output and price which will yield maximum profit and this level of output and price will not be the optimum as realized under perfect competition. But if the same monopoly firm is in the public sector it is possible to envisage a situation where the firm willingly foregoes its objective of profit maximization and fixes a level of production and output which is in accordance with the optimum. However, a public sector enterprise is characterized by a great deal of vices attributed to this form of enterprise and optimum production and policy conditions are seldom realized in these enterprises and many of them run at a loss.

In between the extremes of monopoly and perfect competition we can envisage situations like duopoly,

oligopoly, monopolistic competition and so on. Duopoly is a situation where there are only two firms in the market and oligopoly is a situation where there are a few firms in the market. Monopolistic competition is a situation about which a great deal has been written by economists. This situation is characterized by differentiation of products where a large number of firms produce items which are not exactly similar to each other but closely resemble each other. The mobile hand set market is an example. This is a situation which commonly prevails in the modern corporate sector. Even under monopolistic competition the output of the firm is smaller and prices higher than under perfect competition.

Under conditions other than perfect competition the government generally feels a need to intervene in the market and introduce some kind of regulation over the market forces. However, too much regulation over industry is resisted by the private sector and this is also considered harmful for the healthy growth of private enterprise. Nevertheless, some kind of anti-monopoly and regulatory legislation has been enacted by most governments which is considered to be in the interests of society.

Even under perfect competition the firms operate with the objective of maximization of profits. But it is a unique feature of perfect competition that under it, optimum levels of production and prices are realized. Here the authority of the 'invisible hand' rules supreme and its authority goes unquestioned. But under non-perfect competitive conditions the role of the market forces has to be supplemented by market intervention in the interests of society at large.

CHAPTER V

NATIONAL INCOME

The Gross National Product (GNP) or National Income of a country is a measure of the total volume of economic activity that takes place in the country during the year. The GNP can be estimated by three methods, the product approach, the income approach and the expenditure approach. Theoretically speaking all three approaches should yield the same value of GNP. In practice usually a combination of all three approaches is used to measure the GNP or National Income of a country.

From the product approach point of view it may be said that GNP is the value or sum total of all final production that takes place in a nation during a year. The use of the term 'final' production is significant. The output of certain industries may become inputs in certain other industries. Such products cannot be termed as final products. They are intermediate products whose value is not included in a measure of GNP as it will lead to double counting. For instance, the steel manufactured by one industry may become the input in machinery manufacturing industry. If the value of steel as well as the value of machinery are both included in GNP, it

leads to double counting of the value of steel. Therefore, the value of machinery which is the final product alone is counted in a measure of GNP and the value of steel is excluded. In this way when the value of all final products in an economy is added up, we get the value of GNP in a particular year.

Instead of adding up the values of all final products, GNP can also be estimated by the value added approach. In this approach the value added to total production by all units in an economy is summed up to derive the estimate of GNP. In our example of manufacture of machinery we can imagine three stages of production. First, the mining of iron ore takes place. Then the iron ore is converted into steel. In the third stage, the steel is made into machinery. The value added in the first stage is the full value of iron ore. The value added in the second stage is the full value of steel minus the value of iron ore which went into the production of steel. The value added in the third stage is the value of machinery minus the value of steel that went into the production of machinery. It may be noted that by summing up the value added at each stage of production also we can derive the value of the final product. Total value added in an economy in a year is also a measure of its GNP.

Total value added in production also turns out to be the total income earned by all factors of production. Let us take the example of production of steel. Iron ore is the intermediate product used in the production of steel. The steel manufacturing unit employs the four factors of production to convert iron ore into steel. In the process the unit has to pay rent on its premises, wages to the labor employed, interest on the capital utilized

and, in addition, it earns a certain amount of profit. The value added by the steel manufacturing unit is equal to the sum of the value of rent, wages, and interest paid and profits earned by the unit. Therefore, the GNP or national income can also be estimated as the sum total of all types of income generated in an economy in the form of rent, wages, interest and profit. Thus the second method of estimation of GNP or the income method is to add up the value of all types of income generated in an economy during a year.

Total production in society is either used for consumption by households or by investment by the producer units. (for simplicity we are assuming that there is no accumulation of stocks and no exports and imports. Thus GNP can also be estimated by the expenditure approach as the sum total of all expenditure on final goods and services either for the purpose of final consumption or investment.

All income earned in society is either used for consumption or savings. GNP is also equal to the sum total of consumption and investment. Thus we can derive another accounting identity that total savings in an economy is equal to total investment, both appropriately defined.

Total production in society includes the production of goods and services. The GNP is a measure of total production of goods and services in an economy during a year. Production can take place in the primary sector, in the secondary sector or in the tertiary sector of the economy. The primary sector is generally considered to be comprised of sectors like agriculture, mining,

fisheries, and so on. The secondary sector is primarily the manufacturing sector covering the small scale and large scale industries. The tertiary sector or the service sector is composed of sectors like banking, education, public health, public administration and so on.

The GNP of a country is always expressed in value terms, i.e., in terms of Rupees, Dollars, Yen, SDRs and so on. This is because total production in various sectors of the economy can be added up only when they are expressed in value terms. We cannot directly add up 5 million tons of wheat and 1 million bales of cotton. But when the two items are expressed in value terms as Rs 5000 crores of wheat and Rs 500 crores of cotton the two magnitudes can be summed up as Rs 5700 crores worth of total production.

The GNP or national income of a country goes on increasing from year to year. A country which has a national income of Rs 100 billion in 2014 may have a national income of Rs 108 billion in 2015. This indicates a growth rate in national income of 8 per cent. The rate of growth of national income of a country is an indicator of the progress being made by the country's economy in terms of aggregate production of goods and services.

To give a true picture of economic growth, the growth rate of national income has to be measured in real terms taking account of variation in prices. If the national income of a country grows by 8 per cent in value terms and the general level of prices in the country grows by 3 per cent during the same year, the real growth of the country's economy is only 5 per cent during the year. Thus a distinction can be made between growth of an

economy in nominal terms and real terms. Another way of referring to this distinction is to refer to the former as growth in terms of current prices and the latter as growth rate measured in terms of constant prices.

When a country's economy is growing at a steady rate, it is the most important indication that the health of the economy is sound. When there is an indication that the GNP of a country is likely to record a fall during the current year we can say that a recession is round the corner. A severe recession extending over a period of time may be referred to as a depression. On the other hand, when all indications of an economy are showing a highly positive trend, the economy is experiencing a boom.

The size of the GNP of a country is a measure of the size of the country's economy. Let us say the size of GNP of India was Rs 2000 billion in 2016 and the size of GNP of USA was \$ 400 billion during the same year. To compare the size of GNP of the two countries we have to convert the GNP of the two countries to the same denominator by applying the appropriate exchange rate. Let us say the exchange rate of the Rupee with respect to the US\$ was Rs 50 for a dollar. Using this exchange rate we find that the GNP of India in 2016 was \$40 billion. From this we infer that the in 2016 the US economy was 10 times as large as the Indian economy.

The absolute size of a country's GNP does not give any indication of the standard of living of the population of the country. The best indicator of the standard of living of the population of a country is its per capita

income. The per capita income of a country is the GNP of the country divided by its population. If the per capita income of India is \$ 400 and the per capita income of USA is \$20000 we can say that the standard of living of the people of USA is 50 times that of an average Indian.

CHAPTER VI

POPULATION GROWTH AND UNEMPLOYMENT

The population growth in a country depends on the birth rate and the death rate. The birth rate is the number of live births that take place in a country during a year per thousand population. Similarly, the death rate is the number of deaths that take place per year per thousand population. The difference between the birth rate and the death rate shows the rate of increase of a country's population. If the difference between the birth rate and death rate in a country is 20 during a year it means the for every 1000 population in that country there is an increase in its population by 20 during a year. One-tenth of it or 2 per cent is the rate of growth of the country's population during a year.

Before the advent of the industrial revolution and modern medical practices, the rate of growth of the world population was very low – well below one percent per annum. This was because both birth and death rates were high and the difference between them was low. The death rate was high because of poor medical facilities, a high death rate among infants and children, and the impact of natural calamities like floods,

droughts, famines, etc. the number of children given birth to by a woman was also high during this period as families had to take account of the possibility of many of their children dying before they reached adulthood. Thus the birth rate was also high in the population during the period before the industrial revolution. The high birth and death rates led to slow rate of population growth before the industrial era.

As industrialization proceeded and better medical facilities became available the death rate showed a considerable decline. The gap between the birth rate and the death rate widened and the rate of growth of population increased. This is the second phase of the phenomenon that is referred to as the demographic transition. The second stage is thus characterized by high rates of population growth.

In the third stage of the demographic transition the population growth once again slows down. Thus the population growth rate is low during the first and third stage of the demographic transition. But there is a significant difference between the two stages. During the first stage the gap between birth rate and death rate is low because both birth and death rates are high. During the third stage population growth is low because the birth and death rates are both low.

We have seen that during the second stage the death rate falls and the difference between birth and death rates widens. During the third stage the birth rate also declines and the gap between birth and death rates is again narrowed down. The fall in birth rate occurs during the third stage because as the society advances

and education becomes widespread, families tend to show a preference for having lesser number of children. Moreover, during this stage the death rate among infants and children or the infant mortality rate has also come down and the risk of one's children dying at a young age is considerably reduced as a result of various medical innovations.

The advanced countries of the world passed through the stages of demographic transition much earlier than the poorer countries. The advanced countries have now reached the third stage of low population growth, while the poorer countries of the world are still passing through the second stage of high population growth rates. The higher rate of population growth in the poorer countries act as a destabilizing factor as this leads to lower per capita income growth rates in these countries and consequently, more poverty and inequality in the world.

There is an urgent need in the world today to bring down the rate of population growth especially in the poorer countries. One means of achieving this is to make available birth control facilities to the population at low cost and by educating them about the need for limiting the size of their families. Side by side with this method of direct attempt to control population growth should also proceed other indirect methods which help to create an environment conducive to low population growth rates. Reducing infant mortality rates by vaccinations and other immunization measures and providing better health care for children and lactating mothers is one such method. It has also been found that spread of education, primarily education among females

and also a general improvement in the status of women is conducive to the achievement of low population growth rates.

While attempts are being made to slow down the rate of population growth in developing countries, the developed countries are marching ahead towards the goal of zero population growth. The present objective of population planning at a global level is to stabilize the size of world population at a convenient level where it is not so big as to exert a considerable strain on the limited resources of our planet. From this point of view, it may also become necessary to plan for considerable decline in the size of the world population to the optimum level after the world as a whole has reached a stage of zero population growth.

We have seen that the primary factor leading to an increase in the population of a region is the difference between the birth and death rates in the region concerned. It has, however, to be noted that the population of a region can also increase as a result of the net migration of population to the region. Increase in population of a region as a result of net migration is very common in urban areas and particularly the major cities of the world.

What is important from an economic point of view is not only the absolute size of a country's population but also its distribution as between different age groups. The active members of a country's population, from an economic point of view, is generally the population in the age group 15 to 60 years. The population in this age group is generally considered to be members of the

labor force of the country. The population of a country below 15 years and above 60 years is economically dependent on the active population of the country. The population below 15 years of age are economically dependent on their parents. The population above 60 years of age is also largely dependent on their children or on the savings made by them during their active life time.

Classifying all members of a country's population between 15 and 60 years of age as being economically active is only an approximation. The education of many young members of society continue upto the age of 25 or even beyond and thus their age of entering the labor force is considerably delayed. Many women prefer not to enter the job market at all and many others may choose to enter the job market only after their children are sufficiently grown up. Many physically or mentally handicapped members of the population may never become active job seekers. So also many members of society remain economically active after they have passed sixty years of age. Thus including all members of society between 15 and 60 years of age as members of the labor force of a country is only a good approximation.

The rate of unemployment in a country could be measured as the percentage of the labor force or active job seekers who are unemployed during a period. The problem of unemployment is different for an individual who is seeking a regular employment providing a monthly salary and for an individual who seeks employment on a daily wage basis. There is thus no universally accepted criterion of measuring the rate of unemployment and comparison of unemployment rates of different countries

can lead to many pitfalls. Each country has its own definitions and concepts of measuring unemployment.

In economics we also talk of the concept of disguised unemployment. This is a phenomenon primarily witnessed in developing countries, particularly in the agriculture sector. In under developed countries in the agriculture sector it is not uncommon to employ surplus labor. For instance 12 laborers may be employed on a farm to do the work which can easily be undertaken by 10 laborers. In this case laborers may be considered to be unemployed in disguise. Disguised unemployment is common in backward agricultural economies primarily because wage rates are low and massive general unemployment is prevailing.

Unemployment in a society, particularly educated unemployment, can occur due to inadequate manpower planning. Educational expansion can occur without any regard to the demand for labor with a particular specialization. For instance if the number of graduates in Physics produced by the education system is more than the requirement, there is likely to be unemployment among Physics graduates unless they are able to find jobs in areas which strictly do not belong to them. In manpower planning it is attempted to forecast the demand for labor with different specializations sometime in the future. Once the forecast for demand for labor has been made the intake of our educational system for particular courses of study can be so regulated that the output of the educational system in various fields is not of such a magnitude that produces unemployment. From a long term point of view the expansion of educational

facilities in particular directions should be so regulated that it does lead to large scale unemployment.

In a modern society unemployment can also occur as a result of skills one has acquired becoming obsolete. This type of unemployment is termed as technological or structural unemployment. With the rapid pace of technological progress in our society this type of unemployment is becoming more and more common. To avoid this type of unemployment facilities for in service training or retraining of employees have to be provided by the employers and the employees have to remain alert with regard to updating their skills and acquiring new skills throughout their careers.

In developing societies a primary cause of unemployment is the rapid pace of population growth and expansion of the labor force. Rapid population growth leads to a phenomenon of too many hands chasing too few jobs and resulting in massive unemployment. Thus in developing societies the exercise of population planning has to be integrated with the exercise of manpower planning.

CHAPTER VII

MONEY SUPPLY AND INFLATION

Prices of commodities all over the world show an increasing trend. An overall increasing trend in prices of most commodities is referred to as inflation. The rate of inflation in a country during a specified period is measured in terms of the Price Index. The price index may be referred to as the weighted average of the prices of all commodities during a specified period expressed as a ratio of the prices of all commodities during a base year.

In constructing the price index a large number of representative commodities are to be chosen whose prices are to be inspected from time to time. The choice of representative commodities may be made after a comprehensive consumer survey. The consumer survey will also show the relative importance of various commodities in the consumption basket of an average family or the proportion of an average family's income spent on the purchase of a particular commodity or a particular group of commodities. The proportion of an average family's income spent on a particular commodity will also determine the weight attached to that commodity in calculating the price index.

Once the commodities to be included in the price index are chosen and their weights determined the prices of all such commodities are assigned a value of 100 during a chosen base year. The prices of all commodities in subsequent years are then expressed as a ratio of their prices that prevailed in the base year. For instance if the price of wheat has increased by 20 per cent in 2016, compared to its price in the base year, 2010, the price index of this commodity would be shown to have increased from 100 in 2010 to 120 in 2016. Further if the price of wheat in the year 2020 is double that of what prevailed in 2010 the price index for wheat in the year 2020 would be 200.

The overall index of all prices would also start from a level of 100 in the base year. The overall price index in subsequent years will be the weighted average of price index of individual commodities in the respective years. The term weighted average implies that commodities on which a larger share of income is spent by consumers will be given a higher weightage when determining the average overall index of price level.

In our example the overall index of prices starts with a value of 100 in the base year, 2010. If the overall price index increases to 110 in 2011 it shows that the rate of inflation was 10 per cent during the year or in other words the average increase in prices of all commodities during the year was 10 per cent. Furthermore if by 2020 the overall index of prices touches the level of 200 it means that the average price of all commodities has doubled during a period of 10 years or conversely the real value of the nation's currency has been reduced by half during a period of 10 years. In other words

after a period of 10 years a family has to spend twice the amount of money to buy the same amount of commodities as it purchased 10 years earlier.

Inflation effects different sections of society in different ways. Fixed income groups or salary earners find that with the same amount of income they are able to buy only less and less amount of goods and services as the years pass by. Even when their incomes are partially indexed to the price level, their position tends to worsen over the years. As against fixed income groups the businessmen find that their position generally shows an improvement when there is a moderate increase in prices. This is because the businessmen find that the prices of the commodities they sell tend to increase over a period of time and part of this increase in prices goes to increase their profits or real incomes. The real income of a person is said to have increased when the percentage increase in his nominal or money income is more than the percentage increase in overall prices during the period.

Inflation can act as a serious disincentive to savings. Savings is that portion of a person's income that is not spent during the current period but is set aside to be spent sometime in the future. For instance the salary earners save a certain proportion of their annual incomes every year to be set aside for spending after their retirement. But during a period of inflation the real value of one's savings tends to decline over a period of years. In other words the same amount of money will buy less commodities in the market after 10 years than they do now. The savings of an individual is usually invested in some form of financial assets which yield an

interest income every year. The individual's income so invested does not decline in real value over a period of time unless the rate of interest earned on the savings is lower than the rate of inflation during the period.

A person who invests his savings in property like land and buildings or commodities like gold often tends to gain during a period of inflation. Such a person is at an advantage if the price of the commodity in which he has invested his savings appreciates at a rate more than the rate of inflation.

The causes of inflation or rise in prices are many. One of the factors that causes an increase in the price of a particular commodity is shortage of that commodity. Land for instance is a commodity whose supply is relatively fixed and so its price has a tendency to increase. During a bad agricultural year when the production of certain agricultural commodities shows a decline their prices are likely to increase. The reserves of mineral oils used as a fuel in automobiles are limited. Thus the price of this commodity has a general tendency to increase. Prices of commodities can also go up as a result of artificial scarcity created by the producers or sellers of the commodity.

While the reasons for an increase in the prices of individual commodities may be many, there are primarily three reasons for an overall increase in prices or what may otherwise be termed as inflation. Inflation can occur if there is a shortage in overall production in the economy during a year, the incomes or spending power of individuals remaining unaffected. Inflation can also occur if the incomes or spending power of individuals are

increasing faster than the pace of increase in real production. A third factor generally considered to be the most important cause of inflation is an excessive increase in money supply in the economy. According to the famous 'Quantity Theory of Money' when the pace of increase in money supply far exceeds the pace of general increase in production in the economy there is bound to be an upward pressure on prices leading to an inflationary situation. According to this theory, in the interest of maintaining price stability and overall stability in the economy, it is necessary to maintain the increase in money supply at a rate not far exceeding the rate of increase in production.

The money supply in a country is composed of the coins and currency in circulation plus bank deposits. Some economists prefer to consider time deposits as part of money supply while others prefer to include only the more liquid forms of bank deposits i.e., demand deposits as part of money supply. In any case it is generally observed that there is a close connection between the pace of expansion of all components of money supply including currency and coins, time deposits and the more liquid forms of bank deposits.

In all countries of the world, currency is generally issued by the Central Bank of the country like the Federal Reserve Bank in the USA or the Reserve Bank in India. Usually the government of a country has substantial control over the quantum of issue of currency and coins or primary money supply. It is basically primary money that forms the basis of credit expansion by the banking system or growth in bank deposits. For every increase in primary money supply

there is a multiple (roughly fixed) increase in bank credit or bank deposits. Through the medium of monetary policy the government or Central Bank of a country can exercise considerable control over the expansion in money supply. By appropriate financial management it is possible for a government to limit the pace of expansion of primary money supply. Thus a major instrument of controlling inflation, namely, control over increase in primary money supply rests in the hands of the government or the central bank of a country.

Furthermore there are various instruments of monetary policy whereby the central bank exercises control over the expansion of credit by the banking system. Banks for instance are required to keep with them reserves of primary money in proportion to their quantum of credit expansion. In general the central bank decides what proportion of bank deposits is supposed to be maintained by commercial banks as reserves in the form of primary money. By varying the statutory requirements regarding the proportion of reserves to be maintained by banks the central bank can exercise control over credit expansion by banks or, in other words, expansion of money supply by the commercial banking system. The central banks can also exercise control over growth in primary and overall money supply by manipulating the interest rates and other instruments of monetary policy. Without going into the details of the mechanisms involved it may be pointed out that a lower rate of interest in the economy as a whole is associated with high rates of growth in money supply and higher rates of interest with lower expansion in money supply. Instruments are available with the central bank whereby it can influence the levels

of interest rates prevailing in the economy. By exercising control over growth in money supply the government or central bank of a country can exercise tremendous influence over the rate of inflation in the country and several other important economic variables like the foreign exchange rate of the country's currency.

CHAPTER VIII

PUBLIC FINANCE AND FISCAL POLICY

An interesting distinction between Public and Private Finance has been pointed out by experts in public finance. A private individual first sees his income and then tries to work out his expenditure according to his income. As against this a government first works out its expenditure program and then tries to find the resources for meeting this expenditure. Public finance is that branch of economics which deals with a study of government expenditure and revenue and the impact of government expenditure and revenue raising measures on society.

During the days of Adam Smith, government expenditure was guided by the philosophy 'the best government is one that governs least'. According to this philosophy, the activities of a government were to be restricted to the role of maintaining law and order and protecting the nation against external aggressions. This philosophy was against the present concept of a modern welfare state where the activities of the government have become so widespread in the economy. Apart from fulfilling the role of a police state, a modern government

invests its resources in wide spectrum of activities like infrastructure building, education, providing medical care and implementing other social welfare schemes, besides investing directly in several public sector undertakings. The size of the public administration system has also expanded greatly in all countries of the world.

It is considered progressive if government expenditure is directed towards providing benefit to the weaker sections of society. It is the policy of a modern government to raise resources from the well-to-do and spend them for the benefit of the poor and needy. For instance a great deal of the expenditure of a modern government goes into rural development programs, providing unemployment allowances, pensions and other social security benefits, providing welfare services to the slum dwellers, the homeless, the physically and mentally handicapped and so on. In several instances government expenditure also has an inherent bias in favor of backward and underdeveloped regions of a country. Such a regional bias is justified from the point of view of balanced regional development of the country as a whole.

There are four important sources through which government raises its revenue for meeting the needs of its public expenditure program. The four sources are: taxation, public borrowing, surplus generated by public sector undertakings and deficit financing. In what follows we shall have something to say about each of these four methods of raising public revenue.

The most important source of public revenue is usually taxation. There are two categories of taxes,

namely direct taxes and indirect taxes. Under direct taxes we have income tax, corporate tax, wealth tax, property tax, gift tax, inheritance tax and so on. As against this indirect taxes are primarily taxes on commodities like excise duty, customs duty, sales tax, service tax and so on. In India we are in the process of introducing the Goods and Service Tax (GST) which will replace other indirect taxes like excise duty, sales tax, service tax etc. It has been generally observed that in an advanced country direct taxes are more important from the revenue point of view while in a developing nation indirect taxes are more important.

A tax is considered to be progressive when the burden of the tax is felt more by the well-to-do sections of society. All direct taxes are generally considered to be progressive from this point of view. In the case of income tax certain sections of the population with low incomes are generally exempted from paying this tax. Furthermore, that rate of income tax generally goes up as we move from lower income groups to the higher income groups. In an underdeveloped country the income of a large section of the population is so low that they have to be exempted from the payment of income tax. This is why the importance of income tax is less in a poor country.

While a tax on commodities is not inherently progressive an attempt is usually made to make indirect taxation as a whole progressive. Thus many commodities which are essential items and generally consumed by the poor are exempted from indirect taxes or they are taxed at a low rate. As against this luxury goods and non-essential commodities are taxed at a higher rate.

Many articles like cigarettes, alcohol, etc., whose consumption is harmful from the health point of view are also subject to a heavy dose of indirect taxation. In this way indirect taxes can also be made progressive and also serve a social purpose.

Indirect taxes can also be used to encourage production in certain directions and discourage production of other commodities. By imposing a high rate of indirect tax on cigarettes the production as well as consumption of this commodity can be discouraged. In the same way production of medical equipments and medicines and articles used by the student community can be stimulated by either exempting them from excise duty or subjecting them to lower rates of duty. High custom duties on imports of luxury goods could act as a disincentive to imports of such commodities and also help to save foreign exchange for the country. Furthermore a high rate of customs' duty on certain items could discourage their imports and also encourage the domestic production of these commodities.

A modern government also raises a great deal of revenue through public borrowings. All modern governments issue bonds and other financial instruments to raise resources from the capital market. A major difference between taxation and public borrowing is that while taxes are never to be repaid, resources raised through public borrowing have to be repaid along with a payment of interest. Thus the net resources mobilised by public borrowing is actually the difference between new borrowings and repayment of old loans and interest thereon. The net resources earned by a government through public borrowing is generally positive since the

quantum of new borrowings goes on increasing year after year. While resorting to public borrowings the government has to see that borrowings do not become too much as to create a very heavy repayment burden in the future. The government should also see that while resorting to massive borrowings it is leaving behind sufficient resources of the capital market for investment in the private sector. The government should not borrow too much as to leave the private sector starving for funds.

Another important source of revenue for a modern government are the surpluses or profits earned by the public sector undertakings. It has, however, been observed worldwide that public sector undertakings often do not generate adequate surplus as is expected of them. This shortfall in earning profits may be partly attributed to the managerial deficiencies that are inherent in their very set up and the bureaucratic restrictions imposed upon them. The inadequacy of profits earned by public sector undertakings may also be due to the philosophy that they have been set up to fulfill a social obligation and they are to be run on a non-profit earning basis. This attitude that public sector undertakings should be run on a non-profit earning basis may be justified in special cases where their output is a highly, socially, desirable commodity or service. In such cases their production may even be sold at a loss. But in general in the case of most public sector undertakings there is no justification for not earning adequate profits and thus supplementing the financial resources of the government.

The resources of a government for meeting its - expenditure comes partly from the three sources mentioned

above. The gap between government expenditure and revenue which still remains uncovered is usually filled by deficit financing. Deficit financing essentially means borrowing from the central bank which ultimately results in issue of new currency or increase in primary money supply. Thus this last source of government revenue is potentially highly inflationary. The government should be very cautious in taking resort to heavy doses of deficit financing. Deficit financing of a government should always be kept within prudent limits. Otherwise it can lead to severe inflationary pressures and widespread instability in the entire economy. A theoretical limit to a safe level of deficit financing would be that it should be limited to such an extent where the consequent increase in money supply is in tune with the overall increase in production in the economy.

Government expenditure and deficit financing can also be used as an anti-cyclical weapon. During a period when the economic activities in a country are in a depressed state the economy can be stimulated by a high level of government expenditure and a high level of deficit financing. As against this when the economy is experiencing a boom and an inflationary situation is prevailing the government may try to restrict public expenditure and deficit financing. In an extreme case government can also plan for a surplus budget where the total government revenue is more than government expenditure or deficit financing is negative. The midway approach would be to have a balanced budget where government expenditure is equal to government revenue and the budget deficit or deficit financing is zero. Such anti-cyclical spending is a powerful instrument of Fiscal Policy available with the government to regulate the momentum of economic activity in a country.

CHAPTER IX

THE EXTERNAL SECTOR

The interdependence among the nations of the world has increased considerably. This interdependence is more marked in the economic field than anywhere else. The nations of the world are interlinked together by the bonds of foreign trade, international capital flows and measures taken in the field of international economic cooperation.

For an individual country the most important form of international economic activity is usually foreign trade. Under free trade where there are no restrictions in the field of international trade a country would export those commodities which it can produce at a comparatively lower cost and import such commodities whose domestic cost of production are comparatively high.

Most countries of the world maintain their own currency systems. The currency of India, for instance, is the Rupee and the currency of the United States is the Dollar. There is an exchange rate between the Indian Rupee and the US Dollar at which the two currencies are exchanged with each other. As an example we can

say that the value of one Dollar is equivalent to that of forty Rupees at a particular point of time. In the future if the value of the Dollar appreciates it may become equivalent to that of sixty rupees. Or in the event of a depreciation of the Dollar, it may become equivalent to that of thirty Rupees.

When India exports goods and services to USA it has to express the value of its commodities exported in terms of the US dollar. Thus the price of an Indian product in USA is determined not only by its cost of production in India but also by the exchange rate of the Indian Rupees with respect to the US Dollar. If the Indian Rupee depreciates in the foreign exchange market its products becomes cheaper to foreign purchasers. The depreciation or devaluation of the Indian Rupee can provide a stimulus to the exports from India by making Indian goods cheaper in the international market. Thus devaluation of a country's currency is recommended as a method of giving greater stimulus to a country's exports.

Devaluation while it makes exports cheaper to a foreign buyer also makes imports costlier to the domestic importers. Thus when the Rupee is devalued India will have to pay more Rupees to purchase the same amount of goods from abroad than before. Thus devaluation hurts the interests of the importers and may lead to widespread resistance.

There are several reasons why the exchange rate of a currency fluctuates. Under free trade the exchange rate of a currency is determined by the demand and supply of that currency in the foreign exchange market. But in

several instances as in the case of the Indian Rupee, free exchange of the currency with other currencies of the world was not permitted. Rather the exchange rate of the Rupee was fixed by the Reserve Bank of India from day to day and all exchange of the currency was monitored and regulated by India's central bank. Now such restrictions have been removed and Rupee has been made convertible in the foreign exchange market.

Under the present volatile situation in the international economy, the exchange rate between currencies goes on changing from day to day or even minute to minute. The rate of inflation prevailing in different countries of the world is an important factor leading to fluctuations in the foreign exchange rate. When domestic prices change in one country relative to domestic prices in another country the exchange rate of the two countries' currencies will also normally undergo a change. During a year when prices in India have increased by 10 per cent and prices in USA by 3 per cent it is quite natural to expect a devaluation of the Indian Rupee by about 7 per cent compared to the US Dollar.

The balance of payments position of a country is also closely linked with exchange rate determination. The balance of payments of a country is determined by several factors like foreign trade, international capital movements, and so on. When exports of a country are more than its imports there is a net gain of foreign exchange and the country is said to be having a favorable trade balance. On the other hand when a country's imports are more than its exports there is unfavorable trade balance and a net outgo of foreign exchange takes place.

The balance of payments of a country has many more components besides exports and imports. When direct investment is made by foreign companies in India there is an inflow of foreign exchange. Later on when these companies transfer profits abroad there is an outflow of foreign exchange. When Indian citizens working abroad transfer income to their Indian bank accounts there is an inflow of foreign exchange. When foreign tourists make purchases in India, the country is gaining in foreign exchange. When Indians go abroad either as tourists or as students, the country will lose some of its foreign exchange reserves.

A country is said to have a favorable balance of payments when it is enjoying a net inflow of foreign exchange as a result of its current account transactions. Current account transactions include all payments made because of current purchases of goods and services. The capital account in turn is related to transactions on everything that is not included in the current account. When a country enjoys a surplus or deficit in its current account balance this has to be counteracted by opposite flows on account of capital transactions.

Most of the developing countries in the world are facing an unfavorable balance of payments which is largely due to a surplus of imports over exports. To finance this foreign exchange gap a developing nation takes resort to borrowings from abroad. Such borrowings may be in the form of aid flows from developed countries and international financial institutions or borrowings from the international capital markets. Aid flows from developed countries and multinational financial institutions are either in the form of grants (which is not to be repaid)

or loans at concessional terms of repayment. But borrowings from the commercial markets including the international banking system are at market rates of interest and the other terms of repayment are also stringent. The countries that borrow from international sources should take care to see that their borrowings do not become excessive as it can lead the country into a debt trap. A country which borrows foreign exchange has to repay the loan in foreign exchange. If at the time of repayment the country is not able to mobilize sufficient foreign exchange it may be considered to have fallen into a debt trap.

The three major international institutions established to bring about an order in the field of international trade and capital flows are the WTO (World Trade Organization), the World Bank, and the IMF (International Monetary Fund). In addition, there are a number of regional financial institutions like the Asian Development Bank, the African Development Bank and so on. Another important international economic institution' is the UNCTAD (United Nations Conference on Trade and Development).

The role of the WTO is to bring about orderly relations in the field of international trade. Theoretically a free trade system is considered to be the ideal where the countries of the world gain maximum economic advantage. Free trade ensures that production takes place at the cheapest locations in the world and leads to maximization of production and employment in the world. The objective of the WTO is to take the world forward, as much as possible, to the ideal of free trade. But the WTO has achieved its objectives to only a

limited extent. Major tariff and non-tariff barriers have been imposed by the developed as well as developing countries to protect their economies from competition in the form of imports. Such restrictions on free trade have been justified by the countries imposing them on various grounds. Some of the arguments given in favor of such protectionism or restrictions on trade are: need to protect employment in domestic industries, need for protection to newly established industries from foreign competition, need for conserving foreign exchange, need to restrict imports of non-essential goods, need to promote self-sufficiency, need to protect ailing industries and so on.

As against the limited achievements of the WTO, the World Bank and the IMF have met with more notable achievements. These two sister institutions established jointly at the end of the Bretton Woods Conference in 1944 have a complementary role to play in the field of international finance. The role of the World Bank initially was to provide finance for the reconstruction of the economies affected by the Second World War. After this initial venture at reconstruction, the World Bank subsequently took over the role as the most important financial institution in the world providing finance for world economic development. The World Bank has played an important catalytic role in international economic development programs and its role can be made more effective if its financial capacity can be strengthened by more support from the donor countries.

While the role of the World Bank is to provide development finance, the role of the IMF is to provide balance of payments finance. The IMF comes to the

support of countries facing temporary balance of payments crisis by providing short term finance to tide over temporary difficulties. A controversial issue regarding the IMF lending program is its conditionality. Every country receiving assistance from the IMF is required to sign an agreement to adhere to a predetermined program of economic adjustments and reforms relating to its domestic economy. The conditionalities so imposed by the IMF have given rise to a great deal of controversy relating to their effectiveness and their suitability in relation to the needs of the program implementing countries and compatibility with their political and social environments.

A major innovation of the IMF during its limited history has been the creation of SDRs or Special Drawing Rights. SDR is a form of international currency issued by IMF in favor of all its member countries according to a predetermined formula. Once issued SDRs can be used as a form of foreign exchange to be used to make payments among IMF member countries. However SDRs have not been permitted to be used for day to day market transactions and its role in the world payment system is limited. The conditions regarding the use of SDRs can be liberalized if such an agreement is arrived at by the member countries of the IMF. Properly managed, the SDRs can play a major role in the international economic system. At present, however, its role is limited primarily because the quantum of SDR allocations made so far is limited and also because of the restrictions imposed on its use.

CHAPTER X

POVERTY AND ECONOMIC DEVELOPMENT

The countries of the world can be divided into rich and poor countries or developed and under-developed countries. The under-developed countries or developing countries are those with low per capita incomes and the developed countries or advanced countries or the industrialized countries are those with high per capita incomes. The developing countries of the world are not necessarily those that are poor in natural resources or poor in terms of human resources. There are many developing countries in the world which are well endowed in terms of natural and human resources; still these countries have a very low per capita income and their citizens do not enjoy a high standard of living. On the other side, there are also several industrialized countries of the world that are not very rich in terms of natural resources but meet a large proportion of their requirement of natural resources through imports.

It has been, perhaps, an accident of history that most of the advanced countries, of the world are today located in the northern hemisphere of the world while the world's poorer countries are located by and large

around the equator. Due to these geographic characteristics the developed countries of the world are sometimes referred to as countries of the North while developing countries are referred to as the Southern block. As a result of this distinction the economic conflict between the rich and poor nations is sometimes referred to as the North-South conflict and the negotiations between the two groups of nations is referred to as North-South dialogue. Though it has been an observed reality that the developed nations of the world are concentrated on the northern side of human settlements of our planet and the developing nations are concentrated on the southern side, there is no scientific justification to prove that the backwardness of the developing countries has something to do with their geographic location. Moreover there are indications that these patterns are undergoing a change.

A prominent distinction between developed and developing countries is that the former group of countries are characterized by a high degree of industrialization while the latter group of countries are industrially backward. The greater degree of industrialization in the advanced countries is characterized by a higher proportion of their total production or GNP being accounted for by production in the industrial sector. The industrial sector in the advanced countries is also characterized by a higher degree of technological sophistication and mechanized production and higher productivity of labor. As against this the industries in developing countries are technologically backward, oriented towards processing of primary products and characterized by low labor productivity.

Primary production including agriculture and mining often accounts for a large proportion of the GNP in underdeveloped countries. As against this, the contribution to GNP by the primary sector is much lower in the industrialized countries. Moreover, the productivity of labor is usually much higher in the industrial countries where mechanized farming is the rule and consequently the proportion of the labor force employed in the primary sector is very low in developed countries and very high in the developing countries.

Another distinguishing characteristic of developing countries is their foreign trade composition. A large proportion of the exports of developing countries is accounted for by primary products like agricultural products and minerals while manufactured goods occupy a prominent place in the exports of developed countries. In recent years this characteristic is slowly undergoing a change and manufactured goods are gaining more prominence in the exports of developing countries.

The population of the developing countries being poor, their rate of savings is also low. The low rate of savings result in low rates of investment and capital formation and this once again reinforces low production, productivity and incomes. Thus the developing nations are caught up in a vicious circle of poverty. Development in the backward countries is possible only if they are able to break out of this vicious circle. Countries like India and China have been able to achieve a high rate of savings, yet the impact of high savings rate on development is slow to materialize. Many other developing countries try to overcome their savings

bottleneck by welcoming a larger quantum of foreign aid and direct investment to supplement their domestic resources for investment.

Another major factor retarding the progress of developing countries is the poor quality of human resources in these countries. The population of these countries is abundant in relation to their natural resource endowments and moreover the rates of population growth are also higher in the poorer countries. Thus while human resources are abundant in these countries the quality of human resources leaves much to be desired.

While most of the advanced countries have achieved literacy rates close to 100 per cent, many of the developing countries still have literacy rates below 40 per cent. Thus not only is there a shortage of high level technical manpower in the developing countries, even the basic objective of providing universal literacy remains far from being fulfilled. The scarcity of skilled and professionally qualified manpower is a major hindrance to the progress of developing countries.

Many of the developing countries endowed with abundant natural resources particularly the oil rich nations have development problems of their own. Many of these countries have abundant natural wealth but a shortage of skilled manpower; they are backward in infrastructure development and remain industrially backward. These countries are importing skilled manpower from other developing countries and technology from the industrialized countries to supplement their wealth of financial resources acquired through exports of their mineral wealth for the

purpose of massive infrastructure building activities and other programs of economic development

There are two clearly distinct strategies of economic development followed by the developing countries. One of them is the export led growth strategy and the other is the import substitution strategy. Many of the countries that have adapted an export led growth strategy have made tremendous progress in the matter of economic development. They look forward to the markets of the advanced countries for selling their export products which in most cases consists basically of primary products but in the case of a few selected, countries also include technologically sophisticated manufactured goods. As against this the import substitution strategy lays emphasis on development of the domestic market and gradually replacing imports of manufactured products from developed countries by domestic production of these goods by appropriate strategies of industrial development. A judicious combination of the export led strategy with elements of the import substitution model would be the appropriate policy to follow for most of the developing countries of the world.

A distinction can also be made between balanced and imbalanced development strategies. A country adapting the unbalanced development strategy tries to put in efforts to develop a few selected industries and sectors in which the country has a special advantage. Many of these countries will have to combine their unbalanced development strategy with an export led growth model to achieve best results in the field of economic development. As against this a country adapting a balanced growth strategy tries to diversify its development goals by

simultaneous investment in a wide spectrum of industries and sectors. The overall development that results from this strategy is expected to generate sufficient incomes in the population which is conducive to the growth of a large domestic market and the country can look forward to adapt an import substitution strategy as well. In general it may be pointed out that an unbalanced development model may be more suitable to a smaller country with a limited population while a balanced growth strategy has much to recommend itself in the case of a big country with a large population.

It has been observed that many developing countries adopting planned development programs have achieved substantial progress in raising their levels of per capita incomes. But poverty, malnutrition and illiteracy continue to be a major problem facing these countries. A major drawback of the development strategies adapted by these countries is that their programs have benefited only a certain section of the population which lives primarily in urban areas, are employed in the organized sector and enjoy a high standard of living. As against this a large section of its population, living in rural areas, employed in the low wage unorganized sector or doing manual jobs in the urban areas have benefitted very little from the development programs undertaken by the government. In such countries there is an urgent need to reorient priorities in public sector investment and reorient their plan strategies so that the benefits of development also trickle down to the rural areas, to the unorganized sector and the sections of society that have not benefitted from the development programs so far.

It is not an easy task to measure the extent of poverty that prevails in a country. One such method is to identify a certain level of income as the poverty line and to consider all people having income less than the poverty line as being poor. One way of defining poverty line is to identify it with that level of income which enables an individual to buy sufficient food and other essential goods so that he does not have to suffer from hunger and malnutrition. All individuals who do not earn even this level of income are considered to be living below the poverty line. The proportion of a country's population living below the poverty line may be considered to be a measure of the extent of poverty that prevails in a country. It has been estimated that about 25 per cent of the population of India continues to live below the poverty line even after almost 60 years of planned development after the country gained independence. For an advanced country like USA an entirely different definition of poverty will have to be used to study the problem of poverty; for in that country, hunger and malnutrition is no longer the pressing problem of the economically disadvantaged sections of the population.

The percentage of the population living below the poverty line in a country will depend on the level of per capita income that prevails in the country and also the prevailing levels of economic inequalities. In other words, the level of income as well as the distribution of income are both important in a study of poverty. Lower levels of poverty may prevail in a country with lower per capita income if the income distribution among the country's population is more equal. The income distribution in a country can be considered to be more

equitable when the income disparities between the rich and the poor, the rural and urban population and organized and unorganized labor is less. Thus a program of reducing economic disparities or income disparities should be inbuilt into a program of economic development of a backward country.

The level of poverty that prevails in a poor country is highly correlated with the unemployment situation. Thus a high level of unemployment leads to a high level of poverty. A major attack on rural poverty can be spearheaded through a program of creating jobs in the rural economy. A large section of the labor force in rural areas and the agricultural sector and also construction labor in urban areas are employed on daily wages. The monthly incomes of such labor is highly dependent on the number of days in a month they are able to find jobs. Thus a government program of spending on public works like infrastructure building, providing employment to idle labor at a remunerative wage rate, can have a significant impact on poverty.

The population in rural areas is disadvantaged in many ways. Not only basic infrastructure facilities are not available in rural areas, the rural population is also discriminated against in several ways in the matter of providing institutional credit, in the matter of receiving educational and health facilities, in the matter of availability of electricity, water supply and other public services and so on. Thus all programs of planned development in a poor country should have a special component aimed at improving the conditions of the poor or economically and socially weaker sections in the rural sector.

Another issue that comes up in connection with development of the rural sector and eradication of poverty is the priority that has to be given to the small scale sector. The small scale sector has been given priority in plan programs in several developing countries as this sector is considered to be more labor intensive and thus generates more employment per unit of financial investment. The same amount of money invested in the small scale sector will create more employment opportunities than the same amount invested in the large scale industrial sector. Thus in many developing countries suffering from strong population pressures and unemployment, it would be beneficial if more resources can be channelized into the small scale sector.

The small scale sector is often at a disadvantage with respect to the large scale sector and, therefore, require protection or incentives from the side of the government. It may also be required from the side of the government to provide special facilities like, bank credit on a priority basis, marketing facilities, provision of infrastructure facilities, facilities for promotion of exports from this sector and so on. Properly nurtured and monitored, the small scale sector can make a significant contribution to economic development and poverty eradication. However, blind support for the small scale sector may do more damage than good. There are certain sectors where small units can never become viable. Promoting small scale units in such areas may ultimately result in a high cost economy and the consumers and exports from the country may suffer.

CHAPTER XI

LIMITS TO GROWTH AND URBANIZATION

Ever since the publication of the 'Club of Rome' report on 'Limits to Growth' it has been recognized that there are some constraints to unlimited compound rates of economic growth for the countries of the world. There are many limiting factors in the process of economic growth like limited supply of natural resources (particularly oil), the impact of economic growth on the earth's environment, limitations in the expansion of world food production and so on.

There is one school of thought which believes that these limits will at some time or the other impose a restriction on the pace of world economic expansion. The other school of thought believes that all limits will be overcome as a result of the unlimited possibilities of technological progress. By and large Economic Theory has tried to evade this issue and failed to comment upon it. Side-stepping these issues of vital importance has impoverished Economics and made the tools of Economic Analysis to a large extent irrelevant. A student of Economics soon gets accustomed to the fact that many of the vital issues of his lifetime are ignored by his

subject. An outsider is bound to view the whole issue with greater astonishment.

It has come to be recognized that the mineral resources of our planet are not unlimited. By the end of the century there is bound to be a shortage of many important mineral resources of our planet. The whole issue was dramatized by the oil crisis of the early 1970s. The oil resources of our planet are to a large extent concentrated in the countries of the middle east. It suddenly dawned upon these countries that they virtually enjoyed a monopoly over exportable surplus of this commodity. Oil had also become a sort of an essential commodity for supporting the transport systems all over the world. Living without oil until new technological innovations are brought about would have essentially meant going centuries back in terms of human progress. Furthermore, the planet's resources of oil are not unlimited and it could be easily exhausted within less than half a century. The oil exporting countries of the Gulf region had a high stake in oil. If they let their oil reserves to be depleted at an ever increasing pace, they would one day, before long, find themselves exhausted of this resource and they would have nothing else valuable to export. So they had to make the best use of their oil reserves as long as they are still intact. If they restricted output and increased the price of oil the world will still have to depend on the oil produced by them,

The decision to quadruple the price of oil in the first half of the 1970s was a bold decision which was well-justified from the point of view of the economic interests of the Gulf countries. For the rest of the world it was a

lesson to be learnt. It was a warning that the resources of the planet are limited. Measures had to be taken to conserve the resources of the planet. The immediate problem was to conserve the planet's energy resources. The world learnt a valuable lesson with a major shock, They immediately took measures relating to energy conservation.

The consumption of oil by the transport and other sectors had to be curtailed. The world economy could no longer function taking for granted unlimited supplies of oil and other natural resources.

The high price of oil by itself was a disincentive to the increasing consumption of this vital resource. Besides fuel efficient cars and other energy saving devices became the fashion the world over. Greater attention was directed towards alternative sources of energy like solar energy, wind energy, geothermal energy, energy of the oceans and other such renewable sources of energy. Coal, which was in the process of being forgotten staged a comeback. Nuclear energy was projected as an unlimited source of energy for the world for all time to come, ignoring all its potential environmental hazards. Within nuclear energy, nuclear fusion was highlighted as being both environmentally safe as well as unlimited in supply. The oil crisis of the 1970s also persuaded man to explore more earnestly the possibility of mining the sea bed and, if possible, even exploiting the mineral resources of the objects in space. The world had learned a lesson the hard way.

While on the one side the mineral resources of our planet are limited, the capacity of our planet's environment

to assimilate industrial, transport, agricultural and' other human waste is also limited. Air and water pollution are assuming alarming proportions threatening the very survival of any form of life on the planet earth. The earth's atmosphere and its oceans and other water reserves are no longer considered to be a safe dumping ground for all types of waste generated by man's production activities. Industrial and transport pollution had necessarily to be regulated. Use of synthetic fertilizers and pesticides in agriculture was also polluting the earth's environment and endangering the existence of many living species. It was realized that economic growth and higher levels of GNP was not an unmixed blessing. Economic growth was, also being achieved at a price which economists had for a long time failed to take note of.

The population of the earth was also growing at a fast pace and was adding to the pressures being exerted on the earth's limited resources. At one time it was felt that the only _limit to population growth was exerted by the capacity of the earth to grow food. This constraint still remains very important and vital. But many other important constraints are also now receiving attention. More population means more energy is required to provide a satisfactory standard of living to the earth's inhabitants. In the same way a large population exerts pressure on every other limited resource of the planet and also contributes to greater degradation of the planet's environment.

Economists at one time believed that the greatest evil in the economic system was a shortage of aggregate demand. The argument was that when demand was

high, production would be high, investment would be high, incomes would be high, employment would be high and economic prosperity would prevail. This line of thinking led some people to believe that war and high military spending was a blessing in disguise for the economy, as it would provide a stimulus to overall production in the economy. This line of thinking would be suicidal in the present day world. Today the major economic problem of the day is no longer a shortage of aggregate demand but - a shortage of resources. Every increase in military spending is aggravating the pressure on the planet's finite resources besides adding to the degradation of the environment. Besides being a major political evil, military spending" is also its major economic evil in the world today. There can be no greater economic achievement in the world today than an agreement to divert at least part of the resources being used for military purposes towards provision of foreign aid to the poorer countries.

Just as there are limits to the process of economic growth there are also limits to the process of urbanization. The trend in the present day world is towards greater and greater urbanization leading to over crowded cities, traffic jams on 'major city roads, inhuman living conditions for the poor in the cities and shortage of all types of public services and infrastructural facilities in the urban areas. There is a vital need in the world today to arrest the process of urbanization and initiate a process of de-urbanization.

The net migration to urban areas which has been witnessed from time immemorial has been primarily due to two factors. First of all, the urban areas provided

better living conditions and many facilities which were not available in rural areas. Secondly, under circumstances of unemployment in rural areas the urban areas provided an increasing scope for employment. The process of de-urbanization can be made effective by reversing the two processes mentioned above. First of all, there is a need to provide better living conditions in rural areas and making available all types of essential facilities in rural settlements. With the advent of television, the information and communication revolution, high speed transport. facilities and other technological advances, there is now a better scope than ever before of making life in rural areas more attractive and stimulating. Secondly, efforts could be made for greater employment generation in rural areas and also restricting the scope for employment generation in urban areas There should be a strong and deliberate attempt in this direction if our major cities are not to be choked to death by heavy population pressures. De-urbanization does not only mean a movement from urban areas to rural areas. It should be a continuous process of movement from big cities to larger towns to smaller towns to rural settlements - that is all the way from top to bottom.

Limits to the process of growth and urbanization is a reality to be faced and not to be ignored. The longer the solution is delayed or postponed the more difficult it will become to find a solution. Man will have to live with these twin problems for a long time to come.

CHAPTER XII

ECONOMIC SYSTEMS

The distinction between the capitalist and socialist economic systems is not clear cut. The pure capitalist form of government and the pure socialist form are like two extremes of the economic system's spectrum with a whole range of combinations in varying shades lying in between. The primary characteristics of a capitalist system is the belief in 'private enterprise' and the virtues of the market economy. A pure capitalist may be considered to be one who has complete faith in the 'invisible hand'. A pure capitalist believes in the minimum of government interference over the economic system. He believes that the operation of market forces and the interplay of private enterprises will result in optimum development patterns in the economy. As against this a pure socialist believes in centralized planning where all important decisions regarding production, investment, resource allocation, determination of prices, exports and imports are taken by a centralized planning authority. The pure socialist also has 'little faith' in private enterprise and believes that all economic activity should be controlled and regulated by the government.

As against the two extremes, the proponents of the mixed economic system believe in a certain amount of government intervention and economic planning to regulate and supervise the operation of market forces. It is very difficult to decide where along the spectrum does the capitalist system give way to the socialist system. The distinction between the systems is not clear cut and well defined.

The economic system adapted by a country has to go hand in hand with the political and social systems. For instance under centralized planning it becomes imperative that the political system also has to be centralized in character. Political freedom cannot be achieved without some amount of economic freedom. In the same way the decisions of a centralized planning authority has to be imposed all down the line; a great deal of political freedom has also to be curtailed.

Under centralized planning the government decides what types of goods and services are to be produced, in what quantity, and sold at what prices. In a free enterprise system these decisions are made as a result of operation of market forces. Under mixed economy, the operation of market forces is deliberately influenced by the use of monetary and fiscal policies and other government regulations to achieve what the government considers to be optimum conditions in the economy.

Under free enterprise the demand for a commodity is brought into equilibrium with supply as a result of the operation of the price mechanism. Prices of goods and services fluctuate in the economy bringing about equality of supply and demand. As against this in a

controlled economy the prices are fixed by the government and such administered prices may lead to excess demand over supply of a commodity. When demand exceeds supply many customers will have to go without a commodity which they desire to purchase at the given price or the government will have to introduce a system of rationing where all consumers are permitted to avail of a fixed amount of the commodity.

The proponents of the mixed economic system claim that their system tends to combine the best of both worlds. Freedom of private enterprise is allowed where market forces can produce the best results and government regulation is imposed where market forces can lead to distortions and, sub-optimal economic conditions. In practice it is sometimes observed that a mixed economy is one where the evils, of both system predominate. The opponents of a mixed economic system argue that this system lacks the dynamism of both capitalist and socialist systems. A capitalist system makes progress as a result of free play of private enterprise and their unregulated growth and dynamism. A socialist system progresses as a result of strictly imposed economic discipline and well directed efforts towards economic growth. The dynamic elements of both free enterprise and government regulations are absent in a mixed economy, it has been argued by some.

In the present day world a pure capitalist economy or a pure socialist economy are both non-existent. Looking at it this way, it may appear that all economic systems in the world today are mixed economies. Some amount of free enterprise and some amount of government regulation over economy can be found in all countries

of the world today. The countries of the world today can be classified as capitalist and socialist only in a relative sense.

It has been a long debated issue as to which of the three systems leads to faster economic growth. No conclusive answer can be given to this question as there are fast as well as slow growing economies among countries adopting all three systems. It can also be questioned whether the superiority of the systems is to be evaluated on the basis of the rate of economic progress they, can lead to. It has been observed that many low income countries of the world today have been able to provide a better quality of life to its population than countries with much higher levels of per capita income. Thus the superiority of a country's economic system should be evaluated on the basis of a variety of considerations like pace of economic growth, quality of life enjoyed by its citizens, the pattern of income distribution in the society, the extent of economic and political freedom enjoyed by its citizens and so on.

CHAPTER XIII

ECONOMIC FORECASTING

It is commonplace that everyone has some interest in knowing about his future. In ancient times the system of astrology, palmistry and so on evolved as a result of man's interest in predicting his future. In modern times the methods of forecasting are becoming more scientific and based on broad statistical foundations and computer analysis. The science of forecasting the future development of human society is known as futurology. Futurology deals with the development of mankind in all fields of endeavor including science and technology and its impact on economic social and political developments. In economic forecasting the coverage is restricted primarily to the forecasting of crucial economic variables, but developments in science and technology and social and political developments cannot be totally ignored. Sometimes the role of science and technology is introduced into economic forecasts by means of a single variable, i.e., the pace of technological progress.

Most of the theories in economics can be expressed in a mathematical form. A detailed and serious study of -economics is _today. not possible without a good

knowledge of mathematics. Mathematics gives greater precision to the process of formulation and expression of economic theories. However, it sometimes happens that economists make their economic theories so mathematically sophisticated that one forgets 'the economic rationale behind them and gets lost in a world of figures, formulas and mathematical equations. Thus the use of mathematics in economics has' sometimes been useful and sometimes not.

The fact that economic theories can be expressed in mathematical terms, lends them suitable for statistical analysis. Moreover the value of most of the economic variables like production, prices, rate of economic growth, money supply, revenue from, taxes, level of deficit financing, value of exports and imports. can all be expressed in terms of numbers. This gives rise to a great deal of scope for statistical analysis in economics. The branch of economics dealing with statistical analysis of economic data is known as Econometrics. Today a study of economics at the higher level is considered to be incomplete without a study of Econometrics.

With the aid of Econometrics and Mathematical Economics, economists are able to develop various simplified statistical models relating economic variables which can be used for economic forecasting. For an industry it is useful to forecast what will be the sales of various commodities and services sometime in the future. A government tries to make forecasts of revenue from various taxes in future years. A businessman setting up a factory to manufacture tyres used in motorcars will certainly be interested in knowing the demand for tyres in the future. The demand for tyres in

the future will be related to the demand for new motorcars in the future, sale of motorcars in the recent past, the present and future stock of motorcars in the country, the possibility and scope for export of tyres, likely imports of tyres and motorcars in the' future, likely investment in production of tyres by other manufacturers and so on. With the aid of econometric analysis and the use of statistics available for a large number of economic variables, a rough forecast about the future demand for tyres manufactured by the investing company can be attempted.

A very-special type of economic forecasting is Macro Economic Forecasting. A macro-economic forecast is made possible by the use of macro-economic models developed by econometricians with the aid of computers and a large volume of historical and current data relating to a large number of economic variables. In a macro-economic forecast it is possible to make forecasts simultaneously about a large number of economic variables like future production in different sectors of the economy; future growth rate of the economy, level of prices, money supply, tax revenue etc. in the future, level of incomes, wages and productivity in the future and so on. With the aid of computers it is possible to make forecasts of several hundred economic variables at a time using macro economic models.

Even though very sophisticated and scientific methods are used in macro-economic and other economic - forecasting models, these forecasts do not necessarily turn. out to be accurate. All economic forecasting is subject to. wide ranging errors. But they are certainly of great use in- economic planning and economic policy

formulation. Several great economists with a firm background of economic theory and deep understanding of behavior patterns in the economy are sometimes able to make better forecasts of economic variables than sophisticated econometric models by mere intuitive judgment. But their forecasts are also liable to error as are computer forecasts.

In making economic forecasts, the basic assumption made is that the economy will continue to behave in the future the way it had behaved in the past. Thus by analyzing economic data relating to past years, a forecast can be made of economic behavior in the future. But sometimes it so happens that the basic structure of the economy undergoes such a drastic change, that future economic behavior patterns are drastically different from patterns observed in the past. Such discontinuity can lead to great errors in economic forecasts. One such discontinuity occurred in patterns of economic behavior in industrial countries as a result of the oil crisis during the early 1970s.

The problem of economic forecasting is a science as well as an art. Some models of economic forecasting may yield better results during a particular period and some other models may do better during other times. So also a larger and more sophisticated model does not necessarily lead to better forecasts. A good forecasting model would be one based on thorough scientific analysis but also providing scope for benefitting from sound intuitive judgment of experts.

